

MTE 203 – Advanced Calculus

Week 1 Homework Solutions

Vectors and 3D Analytic Geometry (S.11.1)¹

As mentioned in Lecture 1, the shortest distance between two points (or the length of the line segment joining two points) $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ in 3-D space is given by,

$$\|P_1P_2\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Using your previous knowledge on vector operations (MTE 119) and the concept of distance between points and lines in the 2-D space (MATH 118), solve the following problems in the 3-D space.

Problem 1 [S. 11.1, Prob. 5]:

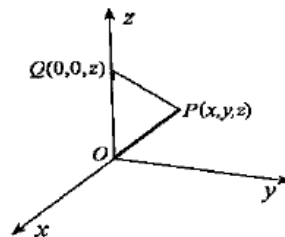
Show that the (unidirected, perpendicular) distances from a point (x, y, z) to the x -, y -, and z -axes are, respectively, $\sqrt{y^2 + z^2}$, $\sqrt{x^2 + z^2}$, $\sqrt{y^2 + x^2}$.

Solution:

If we draw a line from $P(x, y, z)$ perpendicular to the z -axis, the coordinates of Q are $(0, 0, z)$. The length of the perpendicular is

$$\begin{aligned}\|PQ\| &= \sqrt{\|OP\|^2 - \|OQ\|^2} \\ &= \sqrt{x^2 + y^2 + z^2 - z^2} \\ &= \sqrt{x^2 + y^2}.\end{aligned}$$

Similar derivations give distances to the x - and y -axes.



Problem 2 [S. 11.1, Prob. 13]:

- If $(\sqrt{3} - 3, 2 + 2\sqrt{3}, 2\sqrt{3} - 1)$ and $(2\sqrt{3}, 4, \sqrt{3} - 2)$ are two vertices of an equilateral triangle, and if the third vertex lies on the z -axis, find the third vertex coordinates.
- Can you find a third vertex on the x -axis?

¹ This part of the homework is considered a review from past math courses as well as MTE 119 – Statics. It is material that is assumed you already know. For a refresher in vectors, please refer to the MTE 203 - Handout 1 and Sections 11.1 and 11.3 of your textbook.

Solution:

(a) If the third vertex is on the z -axis, its coordinates must be $P(0, 0, z)$. Because this point is equidistant from $Q(\sqrt{3} - 3, 2 + 2\sqrt{3}, 2\sqrt{3} - 1)$ and $R(2\sqrt{3}, 4, \sqrt{3} - 2)$, we can write that $(\sqrt{3} - 3)^2 + (2 + 2\sqrt{3})^2 + (2\sqrt{3} - 1 - z)^2 = (2\sqrt{3})^2 + (4)^2 + (\sqrt{3} - 2 - z)^2$. The solution of this equation is $z = \sqrt{3}$. Since $\|PQ\| = \|QR\| = 4\sqrt{2}$, the triangle is equilateral.

(b) If the third vertex is on the x -axis, its coordinates must be $P(x, 0, 0)$. For P to be equidistant from Q and R , we can write that $(\sqrt{3} - 3 - x)^2 + (2 + 2\sqrt{3})^2 + (2\sqrt{3} - 1)^2 = (2\sqrt{3} - x)^2 + (4)^2 + (\sqrt{3} - 2)^2$. The solution of this equation is $x = -1$. Since $\|PQ\| \neq \|QR\|$, the triangle is isosceles but not equilateral.

Planes and Lines (S.11.5)

Lines in 3D

Find the vector, parametric, and symmetric (if possible) equations for the following straight lines:

Problem 1 [S.11.5,13]:

Find the vector, parametric, and symmetric (if possible) equations for the straight line that passes through the point $(-1, 3, 6)$ and is parallel to the vector $(2, -3, 0)$.

Solution:

A vector equation is $\mathbf{r} = (x, y, z) = (-1, 3, 6) + t(2, -3, 0)$. By equating components we obtain parametric equations $x = -1 + 2t$, $y = 3 - 3t$, $z = 6$. By solving the first and second for t , partial symmetric equations are $\frac{x+1}{2} = \frac{y-3}{-3}$, $z = 6$.

Problem 2 [S.11.5,17]:

Find the vector, parametric, and symmetric (if possible) equations for the straight line that passes through the point $(1, -3, 5)$ and is parallel to the line

$$\frac{x}{5} = \frac{y-2}{3} = \frac{z+4}{-2}$$

Solution:

Since the direction of the given line is the vector $\vec{v} = (5, 3, -2)$, the required line is parallel to $\vec{v}$. Hence, the vector equation of the required line will be given by,

$$\vec{r}(t) = (1, -3, 5) + t(5, 3, -2)$$

By equating the components of the vector equation, we obtain the parametric equations as,

$$x = 1 + 5t$$

$$y = -3 + 3t$$

$$z = 5 - 2t$$

And by solving each one of these parametric equations for t , the symmetric equations are,

$$\frac{x-1}{5} = \frac{y+3}{3} = \frac{z-5}{-2}$$

Problem 3 [S.11.5,19]:

Find the vector, parametric, and symmetric (if possible) equations for the straight line that passes through the point of intersection of the lines,

$$\frac{x-1}{2} = \frac{y+4}{-3} = \frac{z-2}{5}$$

$$\frac{x-1}{2} = \frac{y+4}{3} = \frac{z-2}{4}$$

and is parallel to the line joining the points $(1,3,-2)$ and $(2,-2,1)$

Solution:

By simple observation and using the definition of symmetric representation of a line, the given lines intersect at the point $(1,-4,2)$.

A vector along the line joining the points $(1,3,-2)$ and $(2,-2,1)$ is:

$$(1,3,-2) - (2,-2,1) = (-1,5,-3)$$

The vector equation for the line will be,

$$r(x,y,z) = (1,-4,2) + t(1,-5,3)$$

By equating all components of the vector equation, we can obtain the parametric equations:

$$x = 1 + t, \quad y = -4 - 5t, \quad z = 2 + 3t$$

and by solving each one of these equations for t , the symmetric equations can be written as:

$$x-1 = \frac{y+4}{-5} = \frac{z-2}{3}$$

Planes in 3D

Problem 1 [S.11.5, 9] :

Find the equation of the plane containing the lines

$$3x + 4y = -6, x + 2y + z = 2$$

$$2y + 3z = 19, 3x - 2y - 9z = -58$$

Solution:

Two lines determine a plane only if they are parallel or they intersect. Since the vectors $(3, 4, 0)$ and $(1, 2, 1)$ are normal to the planes $3x + 4y = -6$ and $x + 2y + z = 2$, respectively, a vector along their line of intersection is

$$(3, 4, 0) \times (1, 2, 1) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 0 \\ 1 & 2 & 1 \end{vmatrix} = (4, -3, 2).$$

Similarly, a vector along the other line is

$$(0, 2, 3) \times (3, -2, -9) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 2 & 3 \\ 3 & -2 & -9 \end{vmatrix} = (-12, 9, -6), \text{ as is } (4, -3, 2).$$

Consequently, the two lines are parallel. Since $(-6, 3, 2)$ and $(-11, 8, 1)$ are points on the two lines, $(-6, 3, 2) - (-11, 8, 1) = (5, -5, 1)$ is a vector that also lies in the required plane. A vector normal to this plane is

$$(4, -3, 2) \times (5, -5, 1) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -3 & 2 \\ 5 & -5 & 1 \end{vmatrix} = (7, 6, -5).$$

The equation of the required plane is therefore $0 = (7, 6, -5) \cdot (x + 6, y - 3, z - 2) = 7x + 6y - 5z + 34$.

Problem 2 [S.11.5, 11] :

The acute angle between two intersecting planes is defined as the acute angle between their normals. Find the acute angle between the planes,

$$x - 2y + 4z = 6$$

$$2x + y = z + 4$$

Solution:

The angle between the normals $(1, -2, 4)$ and $(2, 1, -1)$ to the planes is

$$\theta = \cos^{-1} \left[\frac{(1, -2, 4) \cdot (2, 1, -1)}{|(1, -2, 4)| |(2, 1, -1)|} \right] = \cos^{-1} \left(\frac{-4}{\sqrt{21}\sqrt{6}} \right) = 1.935 \text{ radians.}$$

The acute angle between the planes is therefore 1.206 radians.

Problem 3 [S.11.5, 27] :

Verify that the equation of the plane passing through the three points $P_1(x_1, y_1, z_1)$, $P_2(x_2, y_2, z_2)$, and $P_3(x_3, y_3, z_3)$ can be written in the form,

$$\vec{P_1P} \cdot \vec{P_1P_2} \times \vec{P_1P_3} = 0$$

where $P(x, y, z)$ is any point in the plane.

Solution:

Since $\vec{P_1P_2}$ and $\vec{P_1P_3}$ are vectors that lie in the plane, $\vec{P_1P_2} \times \vec{P_1P_3}$ is a vector normal to the plane. Since $\vec{P_1P}$ is a vector in the plane for any point $P(x, y, z)$ in the plane, it follows that $\vec{P_1P} \cdot \vec{P_1P_2} \times \vec{P_1P_3} = 0$, and this must be the equation of the plane.

Distance between Points, Lines and Planes (S.11.6)

Problem 1 [S.11.6, 21]:

Find the distance from the point $(1, 2, -3)$ to the line

$$x = 2(y + 1) = \frac{z - 4}{2}$$

Solution:

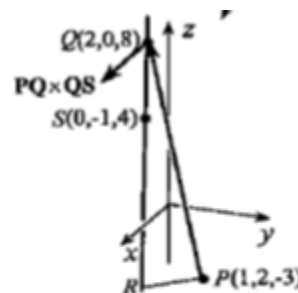
The required distance is the component of $\vec{PQ}$ along $\vec{PR}$. A vector perpendicular to $\vec{PQ}$ and $\vec{QS}$ is

$$\vec{PQ} \times \vec{QS} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 11 \\ -2 & -1 & -4 \end{vmatrix} = (19, -18, -5).$$

A vector in direction $\vec{PR}$ is therefore

$$(\vec{PQ} \times \vec{QS}) \times \vec{SQ} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 19 & -18 & -5 \\ 2 & 1 & 4 \end{vmatrix} = (-67, -86, 55).$$

$$\text{Thus, } d = |\vec{PQ} \cdot \widehat{\vec{PR}}| = \left| (1, -2, 11) \cdot \frac{(-67, -86, 55)}{\sqrt{(-67)^2 + (-86)^2 + (55)^2}} \right| = \frac{710}{\sqrt{14910}}.$$



Alternate Solution:

See the image above for the geometry of the problem. The line can be represented in parametric form as shown below.

$$\begin{aligned}x &= t \\y &= \frac{t}{2} - 1 \\z &= 2t + 4\end{aligned}$$

Then, a point on the line is $S(0, -1, 4)$ when $t = 0$, and the direction vector on the line is $\vec{v} = (2, 1, 4)$.

The required distance is the component of $\overrightarrow{PS}$ along $\overrightarrow{PR}$. $\overrightarrow{PR}$ forms a right angle with the line. The distance can be represented by the following formula using trigonometry,

$$d = |\overrightarrow{PS}| \sin \theta$$

where θ is the angle between $\overrightarrow{PS}$ and $\vec{v}$.

Taking the cross product of $\vec{v}$ and $\overrightarrow{PS}$, taking the magnitude of both sides, and substituting the result above for d gives an alternate form of d .

$$\begin{aligned}\vec{v} \times \overrightarrow{PS} &= |\vec{v}| |\overrightarrow{PS}| \sin \theta \cdot \hat{n} \\|\vec{v} \times \overrightarrow{PS}| &= |\vec{v}| |\overrightarrow{PS}| \sin \theta \\d &= \frac{|\vec{v} \times \overrightarrow{PS}|}{|\vec{v}|}\end{aligned}$$

First find $\overrightarrow{PS}$.

$$\overrightarrow{PS} = (0, -1, 4) - (1, 2, -3) = (-1, -3, 7)$$

Next find $|\vec{v} \times \overrightarrow{PS}|$.

$$|\vec{v} \times \overrightarrow{PS}| = |(2, 1, 4) \times (-1, -3, 7)| = |(19, -18, -5)| = \sqrt{710}$$

Substitute all of the results into the equation to solve for the required distance.

$$\begin{aligned}d &= \frac{|\vec{v} \times \overrightarrow{PS}|}{|\vec{v}|} \\d &= \frac{\sqrt{710}}{\sqrt{21}} \\d &= \frac{710}{\sqrt{14910}}\end{aligned}$$

Problem 2 [S.11.6, 25]:

Find the distance between the lines

$$\frac{x-1}{2} = \frac{y+3}{3} = 4-z$$

and

$$x = -1 + t, \quad y = 2t, \quad z = 3 - 2t$$

Solution:

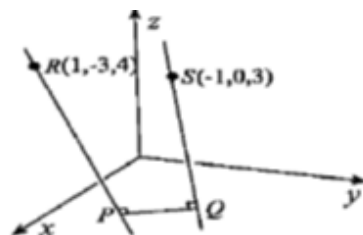
The solution for the distance between two skewed lines has been done in class and can also be found in your textbook, Example 11.24

Following Example 11.24, the distance is the component of $\mathbf{RS}$ along $\mathbf{PQ}$. A vector in direction $\mathbf{PQ}$ is

$$(2, 3, -1) \times (1, 2, -2) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -1 \\ 1 & 2 & -2 \end{vmatrix} = (-4, 3, 1).$$

Consequently,

$$d = |\mathbf{RS} \cdot \widehat{\mathbf{PQ}}| = \left| (-2, 3, -1) \cdot \frac{(-4, 3, 1)}{\sqrt{26}} \right| = \frac{16}{\sqrt{26}}.$$

**Problem 3 [S. 11.6, 41]:**

If a , b , and c (all positive constants) are the intercepts of a plane with the x -, y -, and z -axes, and p is the length of the perpendicular from the origin to the plane, show that

$$\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$$

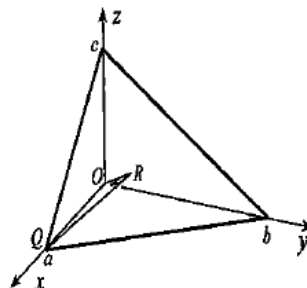
Solution:

The equation of the plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

The distance p from the origin to the plane is the component of $\mathbf{OQ}$ along $\mathbf{OR}$,

$$\begin{aligned} p &= |\mathbf{OQ} \cdot \widehat{\mathbf{OR}}| \\ &= \left| (a, 0, 0) \cdot \frac{(1/a, 1/b, 1/c)}{\sqrt{1/a^2 + 1/b^2 + 1/c^2}} \right| \\ &= \frac{1}{\sqrt{1/a^2 + 1/b^2 + 1/c^2}}. \end{aligned}$$

When this equation is squared and inverted, the required result is obtained.



Extra Practice Problems²

The answers to these problems, along with all even-numbered exercises, are provided at the back of the textbook. Detailed solutions can be found in Trim's Student Solution Manual. If you don't own a copy, a reserved edition is available at the Davis Centre.

1. S. 11.1, Probs. 4, 14, 20
2. S. 11.3, Probs. 8, 20, 30
3. S. 11.5, Probs. 4, 8, 18, 22
4. S. 11.6, Probs. 22, 26, 28, 40

² You are strongly encouraged to try and solve the problems assigned in this homework on your own, before looking at the solutions. These problems are only the minimum necessary to master the material. You are also encouraged to do additional problems from the text for practice (some suggested **extra practice problems** have been listed here).